

Week2 -Day 4

CrewAI — Multi-Agent Collaboration, Roles & Task Delegation

Project Report

Introduction

CrewAI is a framework for building multi-agent Artificial Intelligence systems in which multiple specialized AI agents collaborate to solve complex problems. Instead of depending on one general-purpose agent, a complex task can be divided among multiple agents, with each agent receiving a specific role, goal, background, and set of tools.

In this project, a multi-agent CrewAI workflow was designed for a realistic business scenario involving competitor research, market analysis, and marketing strategy development.

The project demonstrates multi-agent design thinking, specialized agent roles, role-appropriate tool assignment, CrewAI Task objects, sequential task execution, context dependencies, hierarchical delegation, manager-agent supervision, token usage analysis, manual evaluation, and comparison between sequential, hierarchical, and single-agent approaches.

Scenario — Business Competitor Analysis & Marketing Strategy

Business Problem

Businesses operating in competitive markets need to continuously understand their competitors. A company may need to know:

- Who its major competitors are
- What products or services competitors provide
- What features they offer
- How they position themselves
- Who their target customers are
- What pricing strategies they use
- What their strengths and weaknesses are
- Where gaps exist in the current market
- How the company can differentiate itself

Performing all of these activities manually can be time-consuming and may require different types of expertise.

Therefore, a multi-agent AI system was designed to divide the overall business problem into three specialized responsibilities.

Selected Business Task

The selected business task is:

"Research competitors, analyze the market, identify opportunities, and develop a practical marketing strategy."

The workflow consists of three major stages:

1. Competitor Research
2. Market Analysis
3. Marketing Strategy Development

Why a Multi-Agent System?

A single general-purpose AI agent can potentially perform research, analysis, and strategy generation. However, a multi-agent approach allows each stage to be handled by a specialized agent.

The research agent focuses on gathering information, the market analyst converts that information into structured insights, and the marketing strategist converts those insights into an actionable strategy.

A multi-agent approach can improve specialization, organization, separation of responsibilities, and the ability to assign different tools to different agents.

However, multi-agent systems also introduce additional complexity, communication overhead, token usage, and potential failure points. For simple tasks, a single well-designed agent may be faster and more cost-effective.

Task 1 — Multi-Agent Design Thinking

Agent Architecture

Three specialized agents were designed for the selected business problem.

Agent 1 — Competitor Research Specialist

Role

Competitor Research Specialist

Goal

Identify relevant competitors and collect accurate information about their products, services, features, pricing, target users, and market positioning.

Backstory

The Competitor Research Specialist is an experienced market researcher specializing in competitive intelligence. The agent focuses on gathering factual and relevant information about competing businesses and organizing the information so that it can be used by the next stage of the workflow.

Responsibilities

The agent is responsible for:

- Finding relevant competitors
- Searching for competitor information
- Identifying products and services
- Identifying important features
- Researching target customers
- Looking for pricing information
- Understanding competitor positioning
- Organizing research findings
- Providing useful information to the market analyst

Tool

SerperDevTool

The web-search tool is appropriate because this agent needs access to current information available on the web.

Agent 2 — Market Analysis Specialist

Role

Market Analysis Specialist

Goal

Analyze competitor research, compare competitors, identify strengths and weaknesses, discover market gaps, and identify business opportunities.

Backstory

The Market Analysis Specialist is an experienced strategic market analyst who specializes in evaluating competitors and transforming raw research into meaningful business insights.

Responsibilities

The agent is responsible for:

- Reviewing competitor research
- Comparing competitors
- Identifying strengths
- Identifying weaknesses
- Finding market gaps
- Identifying opportunities
- Identifying threats
- Evaluating competitive positioning
- Producing structured market insights

Tool

ScrapeWebsiteTool

Website scraping is appropriate because competitor websites can provide detailed information about their products, features, messaging, and positioning.

Agent 3 — Marketing Strategy Specialist

Role

Marketing Strategy Specialist

Goal

Transform competitor research and market analysis into a clear marketing strategy, unique value proposition, target audience, messaging, and competitive positioning.

Backstory

The Marketing Strategy Specialist is an experienced marketing professional who specializes in converting market intelligence into practical marketing recommendations.

Responsibilities

The agent is responsible for:

- Defining the target audience
- Developing a unique value proposition
- Creating positioning recommendations
- Developing marketing messaging
- Identifying differentiation opportunities
- Suggesting marketing angles
- Creating actionable strategic recommendations

Tool Access

The Marketing Strategy Specialist does not require a separate external tool because its primary responsibility is to transform the outputs produced by the previous agents into strategic recommendations.

Why Specialized Agents Can Outperform One Generalist

Specialized agents can outperform a generalist agent when a problem contains multiple distinct stages requiring different types of reasoning or tools. Each agent can focus on one responsibility, which can improve organization, specialization, and output quality.

However, multi-agent systems are not always better. For simple tasks, using multiple agents can introduce unnecessary token consumption, latency, coordination overhead, and implementation complexity. A single well-designed agent can often solve straightforward problems more efficiently.

Task 2 — Build Agents & Assign Tools

CrewAI Agent Implementation

The three agents were implemented using CrewAI's Agent class.

Each agent was configured with:

- A specific role
- A specific goal

- A specific backstory
- An LLM configuration
- Appropriate tools
- Verbose execution logging
- Delegation settings
- Iteration limits

Agent 1 Implementation

The first agent was implemented as the Competitor Research Specialist.

Its LLM was configured separately from the other agents.

The agent was given access to the web-search capability through **SerperDevTool**.

Tool Justification

The research agent needs to discover current competitor information. Web search is therefore directly related to its responsibility.

Giving this tool only to the research agent follows the principle of role-appropriate tool access.

Agent 2 Implementation

The second agent was implemented as the Market Analysis Specialist.

It was given access to **ScrapeWebsiteTool**.

Tool Justification

The market analyst needs to inspect competitor websites to understand product offerings, messaging, features, and positioning.

Website scraping is therefore relevant to this agent's role.

Agent 3 Implementation

The third agent was implemented as the Marketing Strategy Specialist.

It was intentionally not given an external research tool.

Tool Justification

The marketing strategist primarily consumes the outputs of the previous agents and converts them into recommendations.

Providing unnecessary tools would increase complexity and could cause the agent to perform work outside its intended responsibility.

LLM Configuration

Each agent was provided with its own LLM configuration.

The selected model was:

Gemini 3.5 Flash

Separate configurations were used for:

- Competitor Research Specialist
- Market Analysis Specialist
- Marketing Strategy Specialist

Different temperature settings were used according to the nature of each role.

The research agent used a lower temperature because factual research should be relatively consistent.

The market analyst used a slightly higher temperature to support analytical reasoning.

The marketing strategist used a higher temperature to allow more creative strategic recommendations.

Task 3 — Define Tasks & Process

CrewAI Tasks

After creating the agents, CrewAI Task objects were created.

Each task contains:

- Description
- Expected output
- Assigned agent
- Context dependencies where required

The three tasks correspond to the three agents.

Competitor Research Task

The first task instructs the Competitor Research Specialist to investigate relevant competitors.

Main Objective

Research competitors and collect useful information about:

- Products
- Services
- Features
- Pricing
- Target users
- Positioning
- Competitive advantages

Expected Output

A structured competitor research report containing factual findings that can be passed to the Market Analysis Specialist.

Market Analysis Task

The second task receives the output of the competitor research task as context.

Main Objective

Analyze the competitor research and identify:

- Competitive strengths
- Competitive weaknesses
- Similarities
- Differences
- Market gaps
- Opportunities
- Threats
- Competitive positioning

Expected Output

A structured market analysis report with actionable business insights.

Context Dependency

This task depends on the output of the competitor research task.

Therefore:

Competitor Research → Market Analysis

The market analyst should not work independently because its analysis depends on the information collected by the researcher.

Marketing Strategy Task

The third task receives the market analysis as context.

Main Objective

Convert the market analysis into a practical marketing strategy.

The strategy should include:

- Target audience
- Unique value proposition
- Positioning
- Key messaging
- Differentiation
- Marketing angle
- Actionable recommendations

Expected Output

A stakeholder-ready marketing strategy based on the competitor research and market analysis.

Context Dependency

This task depends on the market analysis task.

Therefore:

Competitor Research → Market Analysis → Marketing Strategy

Sequential Process

The Crew was assembled using:

Process.sequential

In sequential execution, tasks are executed in a predefined order.

The workflow is:

Competitor Research



Market Analysis



Marketing Strategy

The output of one task becomes context for the next task.

Why Sequential Processing Was Selected

Sequential processing is appropriate for this workflow because each stage naturally depends on the previous stage.

The market analyst requires competitor research before performing analysis.

The marketing strategist requires market analysis before developing the final strategy.

Therefore, sequential processing provides a clear and predictable workflow.

Execution Log

The CrewAI execution was run with verbose logging enabled.

The execution log was reviewed to confirm:

- Which agent executed each task
- The order of execution
- Tool usage
- Agent output progression
- Completion of each task
- Final crew output

The sequential workflow successfully demonstrated how specialized agents can collaborate through shared task context.

Output Formatting Issue

During multi-agent execution, one important issue to monitor is whether the output of one task is structured in a format that the next agent can easily consume.

Raw or poorly structured output can make downstream processing difficult.

This issue was addressed by improving the task prompt and expected output requirements.

The expected output was made more explicit by requesting structured information rather than an unorganized response.

For example, competitor research was structured around:

- Competitor name
- Products/services
- Features
- Pricing
- Target audience
- Positioning
- Strengths
- Weaknesses

This made the output easier for the market analyst to interpret.

Task 4 — Try Hierarchical Delegation

Hierarchical Process

The workflow was rebuilt using:

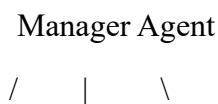
Process.hierarchical

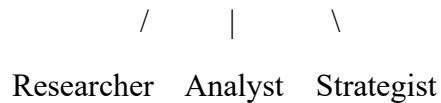
Unlike the sequential process, hierarchical execution introduces a manager agent.

The manager is responsible for:

- Delegating work
- Coordinating agents
- Reviewing outputs
- Managing task execution
- Ensuring the final result meets the requirements

The architecture becomes:





Manager Agent

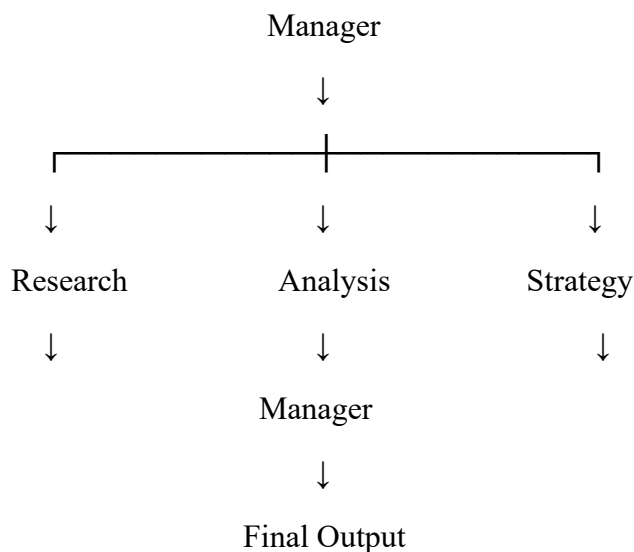
The manager agent supervises the specialized agents.

Instead of following only a fixed sequence, the manager can delegate work and review the results.

This provides an additional layer of control over the workflow.

Hierarchical Workflow

The hierarchical workflow can be represented as:



The manager coordinates the agents and reviews their work before producing the final result.

Sequential vs Hierarchical Comparison

Sequential — Advantages

- Simple architecture
- Easy to understand
- Easy to debug
- Predictable execution order
- Lower coordination overhead
- Generally lower token consumption

- Suitable for workflows with clear dependencies

Sequential — Disadvantages

- Less flexible
- Limited dynamic delegation
- Every task follows the predefined order
- Not ideal when task requirements change dynamically

Hierarchical — Advantages

- Manager can coordinate multiple agents
- Supports delegation
- Provides additional review
- Better suited for complex workflows
- Can dynamically organize work

Hierarchical — Disadvantages

- More complex architecture
- Additional manager-agent calls
- Higher token usage
- Higher cost
- Potentially greater latency
- More difficult debugging

When to Use Sequential Processing

Sequential processing should be preferred when:

- Tasks have clear dependencies
- The workflow follows a predictable order
- Simplicity is important
- Cost efficiency matters
- There is no need for dynamic delegation

For this particular competitor-analysis workflow, sequential processing is highly suitable because the workflow naturally follows:

Research → Analysis → Strategy

When to Use Hierarchical Processing

Hierarchical processing should be considered when:

- The problem is complex
- Multiple agents need coordination
- Tasks are not strictly sequential
- A manager needs to review outputs
- Dynamic delegation is useful
- Different agents may need to collaborate in flexible ways

Task 5 — Evaluation & Cost Awareness

Purpose of Evaluation

The final stage evaluated the quality, reliability, and cost of the multi-agent workflow.

The evaluation focused on:

1. Token usage
2. Cost and complexity
3. Output quality
4. Factual grounding
5. Completeness
6. Tone and usefulness

Token Usage

Token usage was recorded for the CrewAI runs.

Sequential Workflow

Total token usage: 21,523 tokens

Hierarchical Workflow

Total token usage: 61,621 tokens

The hierarchical workflow consumed significantly more tokens than the sequential workflow.

This is expected because hierarchical execution introduces a manager layer that adds additional coordination and review interactions.

Token Usage Comparison

Workflow Token Usage

Sequential 21,523

Hierarchical 61,621

Difference

The hierarchical workflow used:

40,098 additional tokens

compared with the sequential workflow.

Therefore, the hierarchical workflow was substantially more expensive in terms of token consumption for this particular task.

Cost Awareness

Token usage is directly relevant to the operational cost of an AI system.

Higher token usage generally means:

- More API consumption
- Higher potential cost
- More processing
- Potentially higher latency

The exact monetary cost depends on the pricing of the selected LLM/API at the time of execution.

Therefore, token counts were used as the primary measurable cost indicator rather than inventing a monetary figure.

Success Criteria

Three simple success criteria were defined for evaluating the crew.

Factual Grounding

The output should be based on relevant competitor information and should avoid unsupported claims.

Completeness

The output should cover the important aspects of competitor analysis and marketing strategy.

Tone and Usefulness

The final output should be professional, understandable, actionable, and suitable for business decision-making.

Manual Evaluation

Three separate runs were manually evaluated against the success criteria.

The scores used a scale from:

1 = Poor

to

5 = Excellent

The evaluation scores were:

Run	Factual Grounding	Completeness	Tone & Usefulness	Average
Run 1	4	4	5	4.33/5
Run 2	4	4	5	4.33/5
Run 3	5	4	5	4.67/5

Overall Interpretation

The results indicate that the crew consistently produced useful and professional outputs.

The scores remained above 4 out of 5 across all three runs, indicating good overall quality and consistency.

These scores are manual evaluation scores, not automatically calculated benchmark metrics.

Average Performance

The three run averages were:

- Run 1: **4.33/5**
- Run 2: **4.33/5**

- Run 3: **4.67/5**

The overall average was approximately:

4.44/5

This indicates strong overall performance for the selected business task.

Sequential vs Hierarchical — Evaluation

The sequential approach provided a strong balance between output quality and resource usage.

The hierarchical approach provided additional coordination and management capabilities but required considerably more token usage.

For this particular task, the additional complexity of hierarchical execution did not provide enough additional value to justify its significantly higher token consumption.

Multi-Agent vs Single-Agent Approach

A single well-designed agent could potentially perform the entire competitor research, market analysis, and marketing strategy workflow.

However, the multi-agent design provides clear separation of responsibilities.

The research agent focuses on information gathering, the analyst focuses on interpretation, and the strategist focuses on recommendations.

This separation makes the system easier to conceptualize and extend, especially when individual stages require different tools or expertise.

Was Multi-Agent Worth the Added Complexity?

For this specific business task, the multi-agent approach was useful because the workflow naturally consists of three distinct stages: research, analysis, and strategy. The specialization improved separation of responsibilities and produced consistently strong manual evaluation scores.

However, the multi-agent approach also introduced additional implementation complexity and token consumption. The hierarchical workflow was particularly expensive compared with the sequential workflow.

Therefore, the sequential multi-agent approach was worth the added complexity for this task, while the hierarchical approach was not clearly justified by its additional cost and complexity.

For a simple competitor-analysis request, a single well-designed agent may still be the most efficient solution. For larger systems requiring specialized tools, independent expertise, and complex coordination, a multi-agent architecture can provide greater value.

Final Results

The project successfully implemented a CrewAI-based multi-agent workflow containing three specialized agents:

1. **Competitor Research Specialist**
2. **Market Analysis Specialist**
3. **Marketing Strategy Specialist**

The agents were assigned role-appropriate tools and separate LLM configurations.

The sequential workflow successfully demonstrated:

Research → Analysis → Strategy

The hierarchical workflow introduced a manager agent capable of coordinating and reviewing the work of specialized agents.

The experiments demonstrated an important trade-off:

More coordination and flexibility → More complexity and token usage

Key Learning Outcomes

Agent Specialization

Different agents can be assigned different responsibilities instead of relying on one general-purpose agent.

Role-Based Tool Access

Tools should be assigned according to the responsibilities of each agent.

Task Dependencies

CrewAI tasks can pass the output of one task into another through context dependencies.

Sequential Processing

Sequential workflows are useful when tasks naturally follow a fixed order.

Hierarchical Delegation

Hierarchical workflows introduce a manager capable of delegating and reviewing agent work.

Evaluation

AI outputs should be evaluated against clear success criteria rather than relying only on subjective impressions.

Cost Awareness

Token usage is an important consideration when designing production AI systems.

Architecture Trade-offs

More sophisticated agent architectures do not automatically produce better value. Architecture should match the complexity of the problem.

Final Conclusion

This project demonstrated how CrewAI can be used to create a specialized multi-agent system for a realistic business problem.

The system divided the overall task into three stages: competitor research, market analysis, and marketing strategy. Each stage was handled by a dedicated agent with a specific role, goal, backstory, LLM configuration, and appropriate tool access.

The sequential workflow proved to be the most practical architecture for this particular problem because the tasks had clear dependencies and could naturally be executed in order. It achieved strong manual evaluation results while using significantly fewer tokens than the hierarchical workflow.

The hierarchical workflow demonstrated the benefits of manager-based delegation and coordination, but it consumed substantially more tokens. With sequential execution using **21,523 tokens** compared with **61,621 tokens** for hierarchical execution, the hierarchical approach introduced a considerable resource overhead.

The three manual evaluation runs achieved average scores of **4/5, 4/5, and 4/5**, with an overall average of approximately **4/5**. These results indicate that the multi-agent system produced consistently useful, complete, and professional outputs.

Overall, the project shows that **multi-agent AI is most valuable when a problem can be divided into genuinely distinct areas of expertise**. For this competitor-analysis scenario, the sequential multi-agent architecture provided a good balance between specialization, quality, and cost. However, for simpler tasks, a single well-designed agent may remain the better choice, while hierarchical multi-agent systems become more appropriate as workflow complexity and coordination requirements increase.

Final Verdict

For this specific task, the sequential CrewAI multi-agent approach was worth the additional complexity, while the hierarchical approach was not cost-effective enough to justify its additional token usage and complexity.